University of Roehampton London

MSc Data Science

Mathematics and Statistics for Data Science (CMP-L045-0)

Part 1: Applied Problem-Solving Activity

Global Income, Health and Development: A Mathematical Analysis of World Development Indicators, 2015–2022

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Code (GitHub): https://github.com/abshivks-ass/CMP-L045-0-WDI-analysis

Dataset: https://databank.worldbank.org/source/world-development-indicators

Notebook: StudentName.ipynb (renamed to your name before submission)

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Task 1 – Dataset Selection and Justification

Provenance. The data are a raw extract from the World Bank's World Development Indicators (WDI) DataBank [1], an official open-data portal that compiles figures from national statistical offices, the UN Population Division, WHO and the OECD. The extract was downloaded on 1 November 2026 (database last updated 8 October 2026) under a CC BY-4.0 licence. It is not a pre-cleaned teaching file: it arrives in wide format, codes missing values as "..", appends footer text, and mixes 218 economies with 48 regional and income aggregates that would double-count countries if kept.

Variables. Eight annual series for 2015–2022: GDP per capita (current US$), life expectancy at birth (total, female, male), current health expenditure (% of GDP), urban population (%), internet users (%) and fertility rate (births per woman). After reshaping, the panel has 218 economies × 8 years = **1,744 rows** and 8 numeric variables plus country, code and year (Table 1).

Why it is meaningful. The gap in health and opportunity between rich and poor countries is one of the central questions in development policy. This panel lets me test how well the module's tools describe real global inequality: the link between income and longevity, the gender gap in life expectancy, the digital divide, and the COVID-19 shock visible in 2020–2021.

Self-defined criteria. Most analyses use the 2022 cross-section. Countries are split at the 2022 median GDP per capita (US$7,631) into *Higher-income* and *Lower-income* groups (105 each), and thresholds of internet use ≥ 75% and health spending ≥ 6% of GDP define binary attributes.

Missing values in the country-year panel (2015-2022)

	% missing	non-missing n
gdp_pc	2.9200	1,693
life_exp	0	1,744
life_exp_f	0	1,744
life_exp_m	0	1,744
health_exp	11.3500	1,546
urban	0	1,744
internet	12.7900	1,521
fertility	0	1,744

Table 1. Missing values in the 1,744-row panel

Task 2 – Mathematical Application

2.1 Linear Algebra – Vectors

Each 2022 variable is a vector in $\mathbb{R}^n$ with one component per country. Let **f** and **m** be female and male life expectancy (n = 218). Results are in Table 2:

- $\mathbf{f}\cdot\mathbf{m} = \Sigma f_i m_i$ = 1,177,414; $\|\mathbf{f}\| = \sqrt{(\mathbf{f}\cdot\mathbf{f})}$ = 1,124.4; $\|\mathbf{m}\|$ = 1,047.6.
- $\cos\theta = \mathbf{f}\cdot\mathbf{m} / (\|\mathbf{f}\|\|\mathbf{m}\|)$ = 0.99963, an angle of 1.55°.
- The difference vector $\mathbf{g} = \mathbf{f} - \mathbf{m}$ has mean 5.24 years and $\|\mathbf{g}\|/\sqrt{n}$ = 5.57 years (root-mean-square gap); every component is positive (0.39 to 13.22).

The raw cosine is close to 1 for almost any two positive variables, because both vectors point into the positive orthant: it reflects their common level, not association. Centring each vector (subtracting $\bar{x}\mathbf{1}$) turns the cosine into Pearson's correlation, 0.972, so countries where women live long are the same ones where men do. The centred cosine between ln GDP per capita and life expectancy is 0.793. The norm of **g** summarises the female advantage in one number: women outlive men in every economy, typically by about five and a half years.

Task 2.1 Vector operations (2022)

	Value
n (countries)	218
Dot product f.m	1.1774e+06
||f|| (female)	1,124.3791
||m|| (male)	1,047.5521
Cosine(f, m) raw	0.9996
Angle (degrees)	1.5507
Mean gap f-m (years)	5.2384
||f - m||	82.2503
Cosine(f, m) centred	0.9716
n (log GDP, LE)	210
Cosine(logGDP, LE) centred	0.7926

Table 2. Vector operations (notebook output)

2.2 Linear Algebra – Matrices

Data matrix. X is 177 × 6 (2022 complete cases); its columns are ln GDP per capita, life expectancy, health expenditure, urban %, internet % and fertility.

Pre-processing with matrix operations. Columns are centred with $H = I - (1/n)\mathbf{1}\mathbf{1}^T$, which is symmetric and idempotent ($H^2 = H$), and scaled by D^{-1} with $D = \mathrm{diag}(s_1, \ldots, s_6)$, giving $Z = HXD^{-1}$. Min–max scaling is the affine map $(X - \mathbf{1}\min^T)\mathrm{diag}(1/\mathrm{range})$. The code confirms zero means and unit standard deviations.

Gram matrix X^TX. The 6 × 6 Gram matrix holds sums of squares on its diagonal and cross-products elsewhere (Table 3). On raw data it is driven by units: the life-expectancy entry (946,120) mostly reflects the level of about 73 years, not variability. After standardising, $Z^TZ/(n - 1) = R$, the correlation matrix (Figure 1). Internet use correlates 0.86 with ln GDP and −0.84 with fertility, whereas health expenditure is nearly unrelated to the rest ($|r| \leq 0.24$).

X^TX for the raw 2022 data matrix (sums of squares / cross-products)

	log_gdp	life_exp	health_exp	urban	internet	fertility
log_gdp	14,509.0299	1.1659e+05	10,958.4029	1.0098e+05	1.1608e+05	3,645.4139
life_exp	1.1659e+05	9.4612e+05	88,432.2294	8.0833e+05	9.2689e+05	30,193.6899
health_exp	10,958.4029	88,432.2294	10,088.9037	77,237.0208	87,090.3328	2,819.4812
urban	1.0098e+05	8.0833e+05	77,237.0208	7.5807e+05	8.3034e+05	24,427.8314
internet	1.1608e+05	9.2689e+05	87,090.3328	8.3034e+05	9.7774e+05	25,723.8045
fertility	3,645.4139	30,193.6899	2,819.4812	24,427.8314	25,723.8045	1,350.2603

Table 3. X^TX for the raw 2022 data matrix

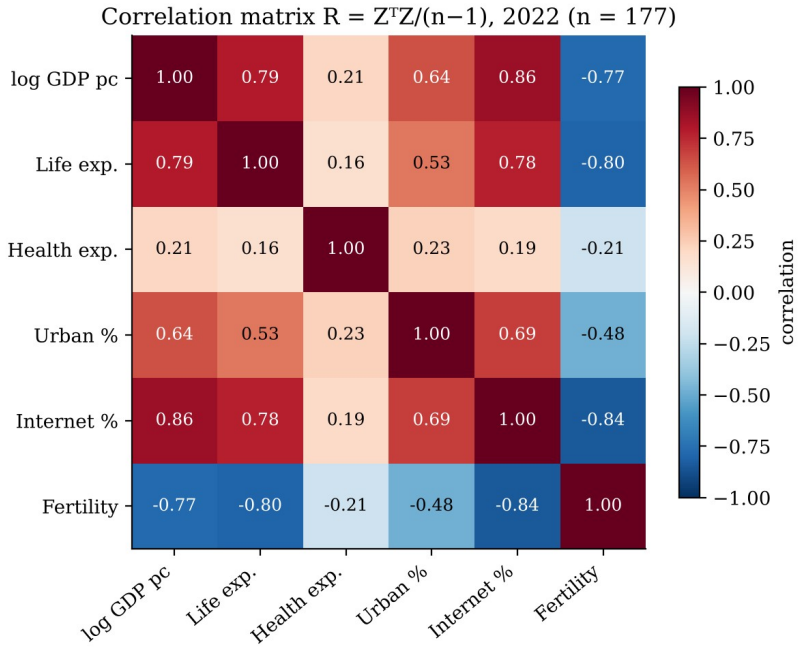


Figure 1. Correlation matrix $R = Z^TZ/(n - 1)$

Determinant. $\det(X^TX) = 7.5 \times 10^{22}$ changes with units, so it is interpreted through R: $\det R = \Pi\lambda_k = 0.0092$ (Table 4). det R lies in [0, 1] and equals 1 only for uncorrelated columns, so a

value below 1% means the six columns occupy a thin region of $\mathbb{R}^6$: their generalised variance is under 1% of that of six independent variables. The eigenvalues agree: $\lambda_1 = 3.96$ carries 66% of the variance (one "development" dimension), $\lambda_6 = 0.10$ is near zero, and the condition number is 39.6. For a pair of variables the 2×2 determinant is $1 - r^2$: 0.265 for ln GDP and internet, so only 26% of their variation is not shared.

In practice. Regression solves the normal equations $\hat{\beta} = (A^TA)^{-1}A^Ty$. Predicting standardised life expectancy from the other five variables gives $R^2 = 0.72$, yet internet use, which correlates 0.78 with life expectancy alone, receives $\beta = 0.05$ because income and fertility already carry its information (VIF = diag $R^{-1} = 6.7$). A small determinant therefore means unstable, hard-to-interpret coefficients, which argues for PCA or dropping variables.

Task 2.2 Determinant and eigen analysis (2022)

	Value
n (rows of X)	177
p (columns)	6
log det(X^TX)	52.6684
det(R)	0.0092
Largest eigenvalue of R	3.9648
Smallest eigenvalue of R	0.1000
Share of variance, 1st PC	0.6608
Condition number of R	39.6310
Max VIF (internet)	6.7120
R^2 (normal equations)	0.7205

Table 4. Determinant and eigenvalue analysis

2.3 Descriptive Statistics and EDA

Statistics use the moment formulas $s^2 = \Sigma(x_i - \bar{x})^2/(n - 1)$, $g_1 = m_3/m_2^{3/2}$ and $g_2 = m_4/m_2^2 - 3$ (excess kurtosis), checked against SciPy (Table 5).

Life expectancy: mean 73.07 < median 74.19, $s = 7.95$, $g_1 = -1.71$, $g_2 = 8.58$. The negative skew is a long left tail: most countries lie between 65 and 85 years, a few far below (Figure 2). The extreme kurtosis comes almost entirely from one value, the Central African Republic (CAF) at 18.8 years in 2022; without it $g_1 = -0.45$ and $g_2 = -0.46$, a mildly skewed, slightly flat distribution. Because the third and fourth moments raise deviations to high powers, one point can dominate them.

GDP per capita: the mean (US$21,110) is almost three times the median (US$7,631); g_1 = 3.05 and g_2 = 12.76 indicate strong right skew and heavy tails, typical of income, which grows multiplicatively. Taking logs gives g_1 = −0.04 and g_2 = −0.84 (Figure 3), close to symmetric, which is why ln GDP is used elsewhere.

Task 2.3 Summary statistics, 2022

	Life expectancy (years)	GDP per capita (US$)	log GDP per capita	LE excl. CAF
n	218	210	210	217
mean	73.0654	21,109.8625	9.0250	73.3154
median	74.1860	7,630.9159	8.9400	74.2110
variance (s²)	63.1713	9.6627e+08	2.1523	49.7770
std dev (s)	7.9480	31,084.8788	1.4671	7.0553
skewness g1	-1.7069	3.0455	-0.0431	-0.4489
excess kurtosis g2	8.5790	12.7625	-0.8423	-0.4641
min	18.8180	301.8323	5.7099	53.9310
max	85.7460	2.2605e+05	12.3285	85.7460

Table 5. Summary statistics, 2022

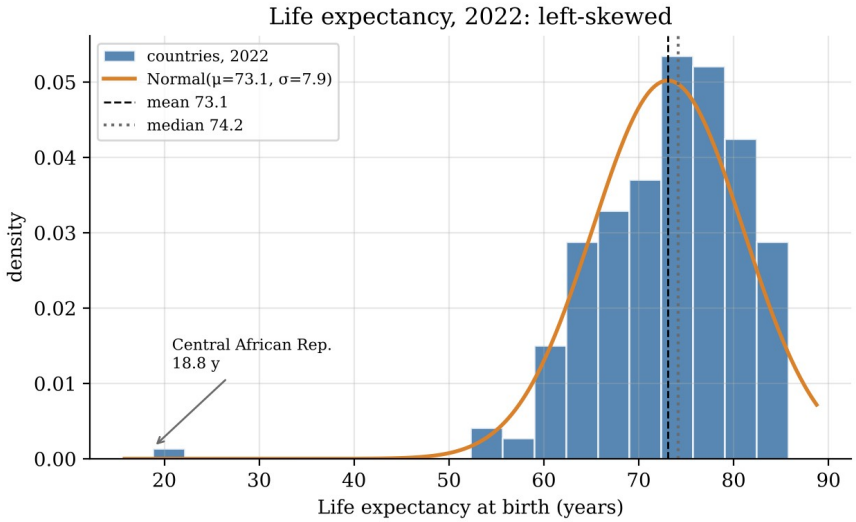


Figure 2. Histogram of life expectancy with fitted normal density

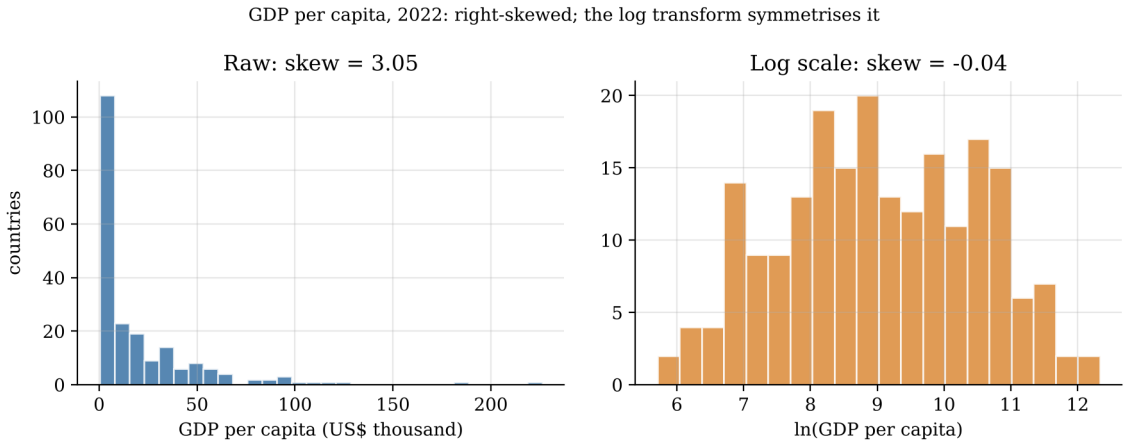


Figure 3. GDP per capita on raw and log scales

Figure 4 shows a near-linear relation between life expectancy and ln GDP: LE = 34.3 + 4.29 ln(GDP), r = 0.79. Doubling income goes with 4.29 ln 2 ≈ 3.0 extra years, a concave, diminishing-returns relation in dollars. CAF is the clear outlier below the line. Figure 5 compares the self-defined groups: the higher-income median is 78.5 years (IQR 6.6) against 67.8 (IQR 8.4); the boxes do not overlap, and lower-income countries are more dispersed. The 1.5 × IQR fence flags one outlier in each group, CAF being the lower-income one.

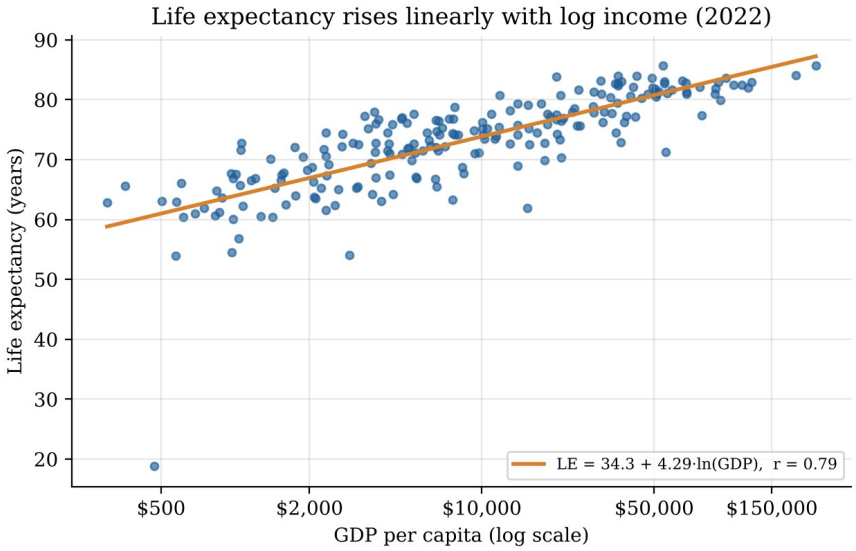


Figure 4. Life expectancy against log GDP per capita

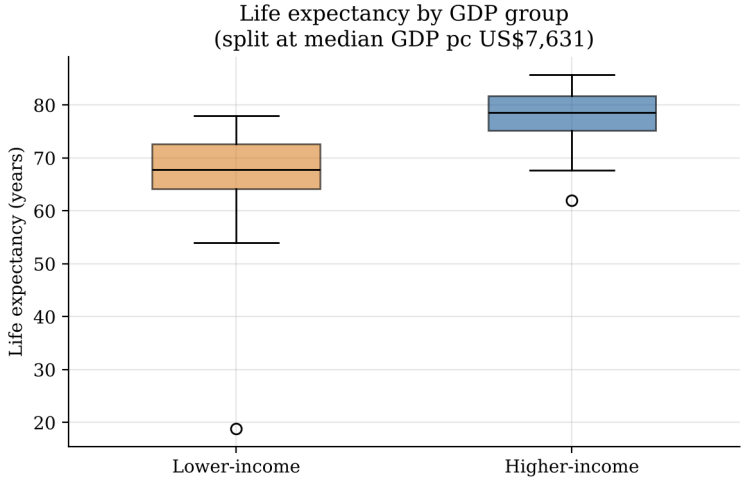


Figure 5. Life expectancy by self-defined income group

2.4 Random Variables and Probability

Discrete variable. K is the number of the seven year-on-year changes (2015–16 to 2021–22) in which a country's life expectancy rose, K ∈ {0, …, 7}. Each change is a Bernoulli trial;

pooling 218 × 7 trials gives $\hat{p} = 0.693$. If trials were independent with constant p, K ~ Bin(7, p), with $P(K = k) = C(7, k)p^k(1 - p)^{7-k}$, E[K] = 7p and Var K = 7p(1 − p).

The empirical PMF peaks at k = 5 (0.362) and k = 6 (0.243) (Table 6, Figure 6). The model matches E[K] = 4.85 (by construction) and the variance (1.495 vs 1.490); $P(K \leq 4)$ is 0.330 observed vs 0.370 modelled. A χ^2 test on pooled cells gives $\chi^2 = 4.49$ (df = 3, p = 0.21), so the binomial is not rejected. Yet the trials are not identically distributed: 81–88% of countries improved in normal years but only 30% and 28% in 2020 and 2021 (COVID-19). The model fits the marginal distribution while its assumptions are false. Under the model, a country improving every year (K = 7) has probability $\hat{p}^7 = 0.077$, close to the 6.4% observed.

Task 2.4 Discrete RV K (N = 218, $\hat{p}$ = 0.693)

	empirical P(K=k)	Binomial(7, $\hat{p}$)
k = 0	0	0.0003
k = 1	0.0092	0.0041
k = 2	0.0275	0.0276
k = 3	0.1101	0.1038
k = 4	0.1835	0.2339
k = 5	0.3624	0.3163
k = 6	0.2431	0.2376
k = 7	0.0642	0.0765
P(K = 7)	0.0642	0.0765
P(K ≤ 4)	0.3303	0.3696
E[K]	4.8486	4.8486
Var[K]	1.4954	1.4902

Table 6. Empirical and binomial probabilities for K

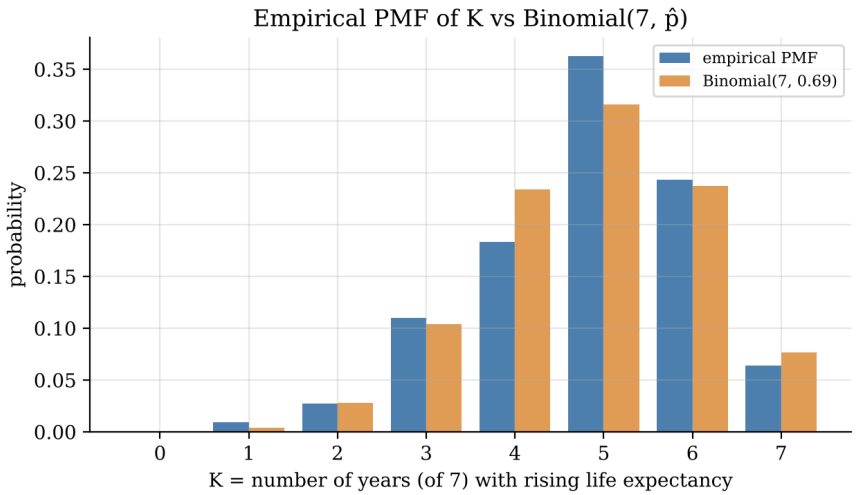


Figure 6. Empirical PMF of K against Binomial(7, 0.693)

Continuous variable. Life expectancy L is modelled as $N(\mu, \sigma^2)$ with $\hat{\mu} = 73.07$ and $\hat{\sigma} = 7.95$. For example, $P(L > 80) = 1 - \Phi((80 - 73.07)/7.95) = 1 - \Phi(0.872) = 0.192$, against 0.216 observed (Table 7). The model puts too little mass on high values and too much in the lower tail (P(L < 60): 0.050 vs 0.028) because the data are left-skewed (Figure 7; Shapiro–Wilk $p < 0.001$). GDP per capita is modelled as log-normal, $\ln G \sim N(9.03, 1.47^2)$, giving P(G > $20,000) = 0.275 vs 0.329 observed: closer in shape (Q-Q r = 0.993), but the log distribution is flatter than normal ($g_2 = -0.84$).

Task 2.4 Continuous RVs: model vs empirical probabilities (2022)

	empirical	model	difference
P(L > 80)	0.2156	0.1915	-0.0241
P(L < 60)	0.0275	0.0501	0.0226
P(65 < L < 75)	0.4037	0.4410	0.0374
P(|L-μ| < σ)	0.6927	0.6827	-0.0100
P(GDP pc > $20,000)	0.3286	0.2747	-0.0539
P(GDP pc < $1,500)	0.1429	0.1216	-0.0212

Table 7. Normal and log-normal model probabilities against observed frequencies

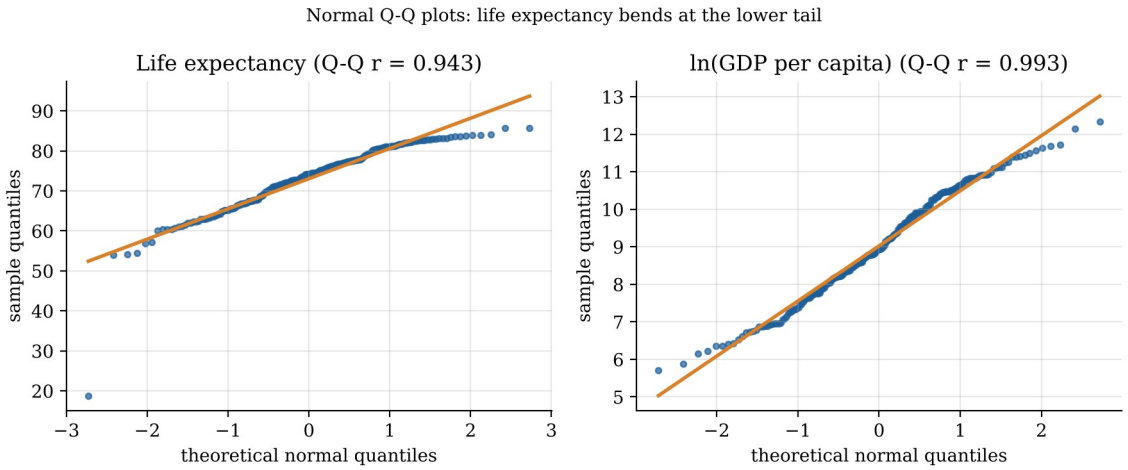


Figure 7. Normal Q-Q plots for life expectancy and ln GDP per capita

2.5 Interval Estimates

All intervals use 2022 data (Table 8, Figure 8).

- **One-sample mean** (life expectancy): $\bar{x} \pm t_{0.975,\,217} s/\sqrt{n} = 73.07 \pm 1.971 \times 0.538 = [72.00, 74.13]$ years. A t interval is used because σ is unknown; with n = 218 the central limit theorem makes it reasonable despite the skew.

- **One-sample proportion** (internet use ≥ 75%): $\hat{p} = 104/182 = 0.571$, SE $= \sqrt{(\hat{p}(1-\hat{p})/n)} = 0.0367$, CI = [0.500, 0.643]. $n\hat{p} = 104$ and $n(1-\hat{p}) = 78$ both exceed 10, and the Wilson interval [0.499, 0.641] agrees.
- **Two-sample mean difference** (life expectancy, higher − lower income, Welch): 78.15 − 67.98 = 10.17, SE $= \sqrt{(s_1^2/n_1 + s_2^2/n_2)} = 0.844$, Welch–Satterthwaite $\nu = 170.4$, $t = 1.974$, CI = [8.51, 11.84] years. Welch is chosen because group variances differ (Figure 5); SciPy returns the same interval.
- **Two-sample proportion difference** (health spending ≥ 6% of GDP): 55/86 = 0.640 vs 43/103 = 0.417, difference 0.222, SE = 0.071, CI = [0.083, 0.361].

Neither difference interval contains 0, so at the 5% level higher-income countries live 8.5 to 11.8 years longer and are more likely to spend at least 6% of GDP on health. The mean interval is narrow (±1.06 years) because n is large, whereas the proportion intervals are wide (±7 to ±14 points) because binary data carry less information per observation. “95%” describes the procedure: across repeated samples, 95% of such intervals would contain the true parameter; it is not a 95% probability for this particular interval.

Task 2.5 95% confidence intervals (2022)

	n	estimate	std. error	critical value	95% lower	95% upper
Mean life expectancy (years)	218	73.0654	0.5383	1.9710	72.0044	74.1264
Proportion internet ≥ 75%	182	0.5714	0.0367	1.9600	0.4995	0.6433
Mean LE: Higher − Lower income	210	10.1730	0.8436	1.9740	8.5078	11.8382
Prop. health exp ≥ 6%: Higher − Lower	189	0.2221	0.0710	1.9600	0.0829	0.3612

Table 8. 95% confidence intervals

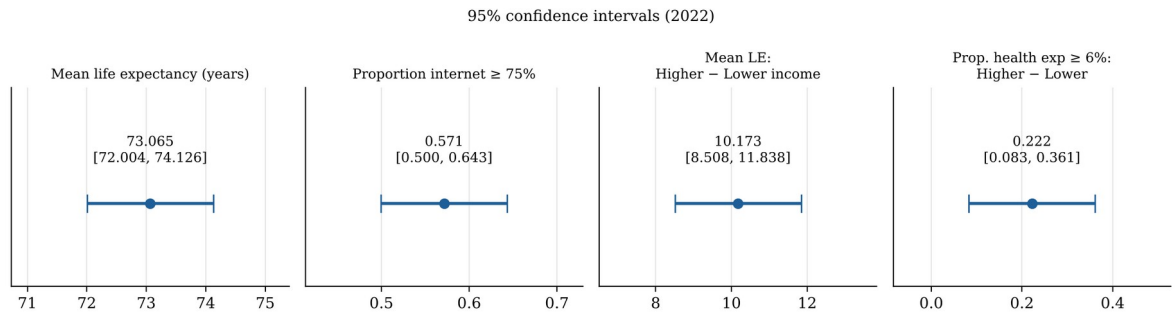


Figure 8. The four 95% confidence intervals

Task 3 – Critical Reflection

Where it aligned. Linear algebra suited the data: centred cosines, Z^TZ and det R exposed one dominant development dimension, and the log transform made GDP nearly symmetric and linear in life expectancy.

Where it broke down. *Randomness:* the 218 economies are essentially the whole population (finite-population correction = 0), so the intervals describe a hypothetical super-population, not sampling error. *Independence:* COVID-19 hit all countries in 2020–21, yet the binomial passed its χ^2 test; a good fit hid a false model. *Normality:* one CAF value moved kurtosis from −0.46 to 8.58, and normal tail probabilities were off by up to 2.4 points. Countries are also weighted equally (Tuvalu counts as much as India), and missing data are not random.

Misuse. A raw cosine of 0.9996 could be read as "identical"; internet use's near-zero coefficient under collinearity as "irrelevant"; a significant interval as proof that income causes longevity. The median split is arbitrary, and another threshold would change the group results. Safer practice is to report robust statistics beside moments, test sensitivity to outliers and thresholds, and state which population an interval refers to.

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Declaration of Generative AI Use

Generative AI (Claude, Anthropic [10]) was used to assist with drafting the Python code, producing figures and drafting parts of this report. All code was run, all results were checked against the notebook outputs, and the analysis and conclusions were reviewed and edited by the author, in line with the University’s guidance on referencing AI.